Semiconductor Properties

1.1 Introduction to Semiconductors

In terms of electrical conductivity, i.e. the ability to conduct current, all materials can be classified under three categories, namely, conductors, insulators and semiconductors. A conductor allows current to flow through it, i.e. it has high electrical conductivity. On the other hand, an insulator tends to block the flow of current. In between these two extremes, we have the semiconductors. Numerically, the **conductivity** of a material, denoted by the symbol σ is the reciprocal of its **resistivity**, denoted by the symbol ρ. Generally for conductors, $\rho < 10^{-3}$ ohm-cm, while for insulators, $\rho > 10^{8}$ ohm-cm at room temperature. In its pure form, silicon, the most widely used semiconductor has a resistivity of 2.31×10^{5} ohm-cm. at 300K. However, by introducing a very small amount of impurities (few parts per million) in silicon, it is possible to reduce this value by several orders of magnitude. This unique property of semiconductors has fuelled intense research in their studies and has culminated in the stupendous success of semiconductor devices and integrated circuits. In fact, it will not be an exaggeration to say that the "Electronic Revolution" of twentieth century is largely due to the ubiquitous semiconductor devices, which are to be found in almost all our daily appliances as well as in sophisticated instruments. It is therefore extremely important to understand the properties of semiconductor materials, as they hold the key to the performance of semiconductor devices.

Elemental semiconductors generally belong to the Group IV in the periodic table. Certain compounds can also be classified as semiconductors. Why is it that some materials are semiconductors, while others are metals and insulators? For a simple

explanation, let us start with the definition of conductivity in a solid, and also look at the factors which affect conductivity. For that, we shall start with Ohm's Law, which states that the current (I) through a resistance (R) due to an applied voltage (V) is given by

$$I = \frac{V}{R} \quad \text{.....(1.1)}$$

Now let us consider the resistance R to be a solid block of homogeneous material having a cross-sectional area 'A' and length 'L' as shown in Fig. 1.1. The electric field (ε) and the current density (J) can be expressed as $\varepsilon = (V/L)$ and $J = (I/A)$. Therefore eqn. (1.1) can be rewritten as

$$J = \frac{\varepsilon L}{RA} \quad \text{.....(1.2)}$$

The resistance of the block is related to the resisivity (ρ) of the material through the well known relation $R = \rho\ (L/A)$. Using this relation in eqn.(1.2) and noting that the conductivity (σ) is the reciprocal of resistivity (ρ), we have

$$J = \frac{\varepsilon}{\rho} = \sigma\varepsilon \quad \text{.....(1.3)}$$

The above equation is another form of Ohm's Law. The important difference between eqns. (1.1) and (1.3) is that while R in eqn.(1.1) depends on the shape and size of the resistance block, ρ and σ in eqn.(1.3) are intrinsic properties of the material.

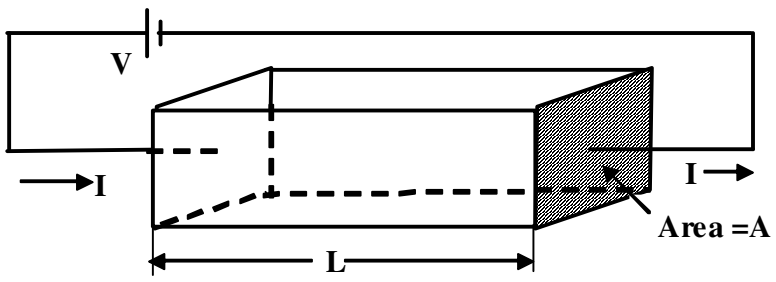


Fig. 1.1 A resistance block of cross-sectional area 'A' and length 'L'

Now let us assume that there are 'N' charge carriers uniformly distributed in the resistance block shown in Fig.1.1. (A charge carrier is a charged particle which can move under the influence of an applied electric field. It can be an electron, which carries negative charge. In the case of semiconductors, we also have a positive charge carrier, called a hole. We shall discuss the concept of a hole later.) Also, let us assume that the charge carriers move under the influence of the applied electric field with a velocity 'v', so that the time (t) required for them to move from one end of the block to the other is given by t = L/v. Thus the number of carriers flowing through a side of the block of area 'A' in time 't' is 'N'. Since the flow of current is continuous, it can be easily understood that the number of charge carriers which pass through any cross-section of the block per unit time is N/t. Hence, the current in amperes, which is defined as the total charge per second passing through any area is given by

$$I = \frac{qN}{t} = \frac{qNv}{L} \quad \text{……..(1.4)}$$

where q is the charge of each carrier. For example, $q = 1.6 \times 10^{-19}$ C for an electron. The current density is therefore given by

$$J = \frac{I}{A} = \frac{qNv}{LA} = qnv \quad \text{…….(1.5)}$$

where n = N/(LA) is the concentration of charge carriers per unit volume. Now comparing eqns. (1.3) and (1.5) we see that

$$\sigma = \frac{qnv}{\varepsilon} \quad \text{…….(1.6)}$$

It has been observed experimentally that at low electric fields, the average velocity of the charge carriers is proportional to the electric field, so that we may write

$$v = \mu\varepsilon \quad \text{…….(1.7)}$$

The proportionality constant μ is referred to as the mobility of the carrier, and its value depends on the material through which it is traveling. For example, the mobility of electrons in pure silicon is about 1500 cm^2/V-sec, while in germanium it is about 3900 cm^2/V-sec. Now substituting eqn.(1.7) in eqn.(1.6), we have

$$\sigma = qn\mu \quad \ldots\ldots\ldots\ldots (1.8)$$

Thus we see that the conductivity of a solid material primarily depends on two factors, viz. (a) the concentration of available carriers, and (b) the mobility of these carriers. We shall discuss the concept of mobility and its dependence on various factors in more detail later in this chapter. However, it may be pointed out that the values of mobility in different materials vary only over a few orders of magnitude, while the carrier concentrations can vary by many orders of magnitude in various material systems. Thus it is the value of 'n' in eqn. (1.8) which primarily determines whether a material is a conductor, a semiconductor or an insulator.

Although in the preceding discussion we have referred to an electron as a charge carrier, we must now point out that all electrons are not charge carriers! We know that an element contains 6.022 x 10^{23} atoms/mole (Avogadro number). Considering this, most materials will have about 10^{23} electrons per cm^3. This is an enormously large number. We shall now explain the reason for the number of electrons participating in electrical conduction to be much fewer. In a simple model of an atom, there is a positively charged nucleus and there are electrons in various orbits around the nucleus. The electrons in the outermost shell are referred to as the valence electrons, while the others which are closer to the nucleus are called the core electrons. These core electrons are strongly bound by electrostatic forces to the nucleus and cannot move under the influence of an external

electric field. It is only the relatively loosely bound valence electrons which can move and participate in current flow. However, even if we consider only the valence electrons, the conductivity obtained using eqn. (1.8) would be extremely large. Also, it cannot be explained why the conductivity is so widely different in carbon, silicon, germanium and tin, which all belong to Group IV in the periodic table and hence have the same number of valence electrons per atom.

To understand this phenomenon, we must consider that we do not have isolated atoms in a solid. The valence electrons actually participate in chemical bonds which hold the solid together. When a valence electron takes part in a chemical bond, it cannot move and hence cannot contribute to current flow. At very low temperatures, almost all bonds are intact. However, with increase in temperature, the electrons gain thermal energy, resulting in bonds being broken and electrons being set 'free'. These free electrons then become charge carriers and can conduct electricity.

There are three types of bonds in solids: ionic, covalent and metallic. An ionic bond is formed between an electronegative and an electropositive element, as in NaCl. These bonds are usually very strong and difficult to break. Hence most materials having ionic bonds are insulators at room temperature. On the other hand, the bonds in metals are formed between electropositive elements. These bonds are very weak and the valence electrons move almost freely throughout the metal. The large number of 'free' electrons results in high conductivity. The covalent bond is formed by sharing of electrons between similar atoms. This bond is of particular interest to us since the Group IV elements form this type of bond. These elements have four valence electrons and these are shared with their four neighbours to form a stable configuration. A two-dimensional schematic

representation of the silicon lattice with the covalent bonds is shown in Fig.1.2. A covalent bond can be quite strong and hence covalently bonded materials are generally quite poor conductors at room temperature. The strength of the covalent bond also depends on the size of the individual atoms. For small atoms like carbon, the valence electrons are tightly bound, while for the larger atoms, like germanium and tin, the bonds are more easily broken. Hence at room temperature, the conductivity of the Group IV elements increases with increase in atomic number due to the larger availability of charge carriers. In the case of the semiconductors, silicon and germanium, their conductivities are more than diamond, which is an insulator, and less than tin, which is a conductor. An interesting effect in semiconductors is the increase in conductivity with increase in temperature. As temperature is raised, the electrons gain energy, resulting in more bonds being broken and free carriers being generated. In the case of metals, however, almost all the valence electrons are already ‘free’ at room temperature. Hence there is no further increase in the number of charge carriers with increase in temperature. Their conductivity in fact reduces at high temperatures due to a fall in mobility of the carriers.

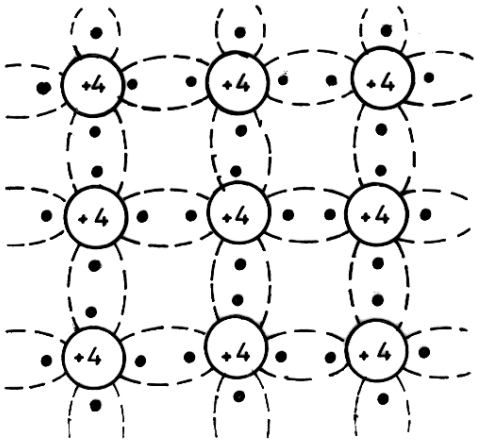


Fig. 1.2 Schematic two-dimensional representation of silicon lattice. The black dots represent valence electrons which participate in bonds.

In addition to the semiconductor elements, some compound semiconductors are now being used for various applications. These semiconductors are usually compounds formed from the elements in Group III and Group V in the Periodic Table. Here again, eight valence electrons are shared between neighbouring atoms. The bonding has a covalent character, but it is partially ionic as well. As a result, the valence electrons are bound to the atoms more strongly than the corresponding Group IV semiconductors. The compound semiconductors, GaAs and InP have lower conductivities than Si. The conductivity increases for compounds of the heavier atoms, e.g. InSb, while compounds of the lighter atoms like BN are good insulators.

We shall now introduce the presence of a positively charged carrier in semiconductors, called a **hole**. When a bond is broken, releasing a free electron, as shown in Fig. 1.3(a), it also leaves behind a vacant site, called a hole, which can easily accept an electron. On application of an electric field, a neighbouring valence electron may move to occupy this hole, thus leaving behind a hole in its original position, as shown in Fig. 1.3(b). Hence the hole has effectively moved in a direction opposite to the direction of movement of the electron. So far as the flow of current is concerned, the hole therefore behaves like a positively charged particle with a charge equal in magnitude to the electronic charge. We thus see that the breaking of a bond results in the creation of an **electron-hole pair (EHP)**. Since they are oppositely charged, and they move in opposite directions under the influence of an electric field, the total current is the sum of the contributions due to the flow of electrons and holes.

It must however be emphasized that this simple picture of a hole as depicted in Fig.1.3 has a number of serious limitations as neither a hole nor an electron is a particle

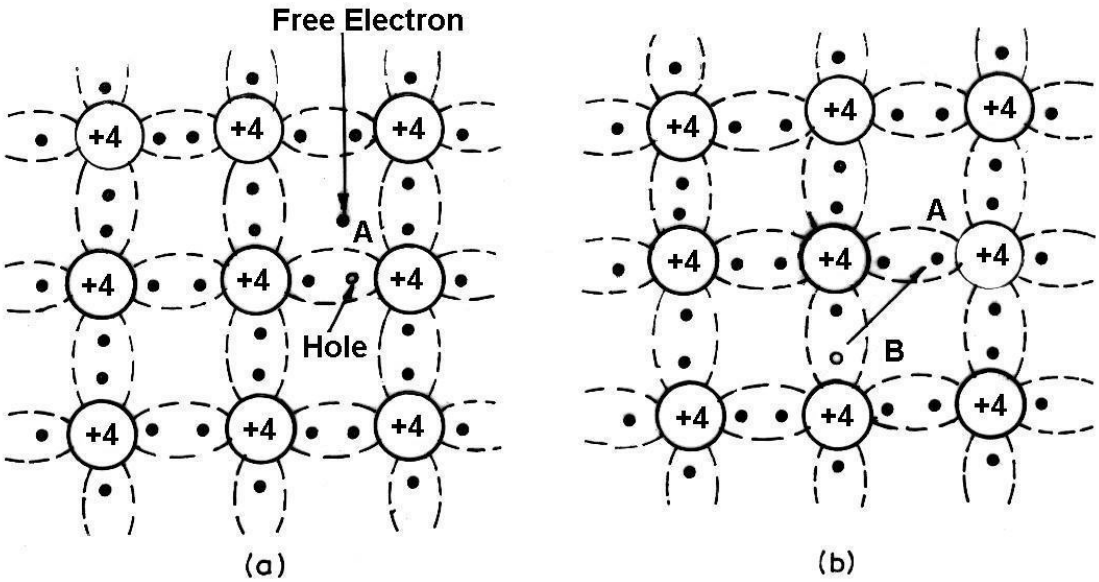


Fig. 1.3(a) A broken bond in position A, resulting in an EHP; (b) A valence electron from position B moves to position A: effectively, the hole moves from A to B

in the classical sense. However, this elementary model helps one to visualize the creation of electrons and holes by thermal generation. We can thus consider holes to be physical entities whose movement constitutes a flow of current.

In addition to the thermally generated intrinsic carriers, it is possible to create carriers in the semiconductors by deliberately introducing impurities in the lattice. This is called **doping**. For example, let us consider the introduction of a Group V element, such as phosphorus in silicon. In terms of the covalent bond formation, the situation can be explained in the following manner. A group V element has five valence electrons, four of which are used to form the covalent bonds. With very little energy, much less than what is required to break a covalent bond, the fifth electron can be made to move freely inside

the crystal. This electron has thus been “donated” to take part in electrical conduction. Such a Group V impurity is called a **donor** as it donates electrons. Analogously, a group III element has only three valence electrons. So there is a hole at the fourth covalent bond site. In order to complete this bond, the Group III element can “accept” one more electron from some other covalent bond site. A Group III impurity is therefore called an **acceptor** as it accepts electrons. In the process, a hole is created at the original electron position. Figs.1.4 (a) and (b) represent these two situations schematically.

A semiconductor in its pure form is called an **intrinsic semiconductor**. In an intrinsic semiconductor, the number of free electrons is equal to the number of holes, as they are generated simultaneously by the breaking of bonds. A doped semiconductor is called an **extrinsic semiconductor**. If the dopant in an extrinsic semiconductor is a donor, free electrons are created without simultaneous creation of holes. Consequently, the number of electrons will be more than the number of holes and the semiconductor is said to be n-type (n stands for **n**egatively charged electrons). Alternately, if the dopant is an acceptor, there will be more holes than electrons and the semiconductor is said to be p-type (p stands for **p**ositively charged holes).

In this Section, we have discussed some of the properties of semiconductors and some other materials using the chemical bond model. However, this is not sufficient to describe the operation of a semiconductor device. A far more powerful model for understanding and describing the operation of semiconductor devices is the Energy Band diagram, which will be presented in the next Section. Subsequently we shall take another look at all the concepts discussed in this section in the light of the Energy Band diagram.

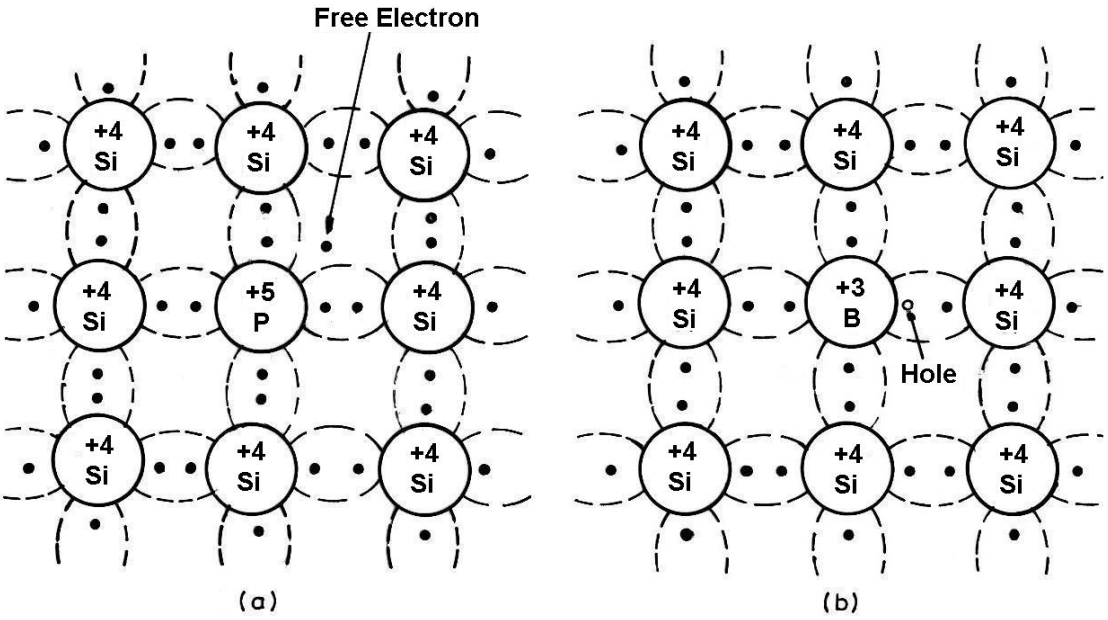


Fig. 1.4 Schematic representation for (a) creation of free electron in n-type Si due to phosphorus (donor) doping and (b) creation of hole in p-type Si due to boron (acceptor) doping

1.2 Energy bands in a solid

1.2.1 Splitting of discrete energy levels into bands

The transport of charge in a solid depends not only on the properties of the electrons but also on the arrangement of atoms in a solid. Solids can be **single crystals**, where the atoms are arranged in a perfectly regular periodic structure in all three dimensions called the lattice**; polycrystalline**, where there are many small regions of single crystal material called the grains; and **amorphous**, where there is no periodic structure. Semiconductors are a particular group of solids at the operating range of temperature. The semiconductors that we are going to study in this book are single crystals. As already mentioned, single crystals consist of a space array of atoms or molecules (or strictly speaking ions) built up by regular repetition of some basic structural unit in three dimensions. This structural unit is called the **unit cell.** Silicon, GaAs and most other semiconductors have a Diamond or

Zincblende lattice structure. In this structure each atom is surrounded by four of its nearest neighbours. Thus the electrons in the outer orbit of one atom feel the influence of neighbouring atoms. Hence the discrete energy levels for a single free atom (as in a gas where the atoms are sufficiently far apart so as not to exert any influence on one another) do not apply to the same atom in a crystal.

As isolated atoms are brought closer, various interactions occur between neighbouring atoms. The forces of attraction and repulsion between atoms find a balance at the proper inter-atomic spacing for the crystal. In the process, important changes occur in the electron energy level configurations, which are responsible for various electrical properties of solids.

Qualitatively, we can say that as atoms are brought closer to each other, the application of **Pauli Exclusion Principle** becomes important. When two atoms are completely isolated from each other (i.e. they are far apart) so that there is no interaction of electron wave functions, they can have identical electronic structure. As the spacing between atoms decrease, the wave functions begin to overlap. The exclusion principle dictates that no two electrons in a given interacting system can occupy the same quantum state. Thus the discrete energy levels of the isolated atoms must split into new levels belonging to the pair rather than to the individual atoms. In a solid, where many atoms are brought close together, the split energy levels form a large number of discrete but closely spaced energy levels called **Energy Bands**.

Let us consider the imaginary formation of a diamond crystal from isolated carbon atoms. Carbon has six electrons in $1s^2\ 2s^2\ 2p^2$ configuration**.** Consider a system of 'n' equidistant carbon atoms. Each atom has two 1s states, two 2s states and six 2p states.

So for n carbon atoms, there will be 2n states of 1s type (all are filled), 2n states of 2s type (all are filled) and 6n states of 2p type (of which 2n are filled and 4n are empty). When the inter-atomic spacing is large, the energy levels of these sub-shells are isolated levels. However, as the atoms are brought closer, these energy levels split into bands, beginning with the outermost sub-shell, as shown in Fig.1.5. As the 2s and 2p bands grow, they first merge into a single band comprising of 8n states of which 4n are occupied. As the distance between atoms approach the equilibrium inter-atomic spacing of diamond, this single band splits into two bands separated by an **Energy Gap (E_g)**. The lower band contains 4n states, all of which are occupied at 0K. The upper band also contains 4n states, but they are empty at 0K. The energy gap contains no available states for the electrons to occupy and is hence referred to as the "forbidden gap". The lower band is called the **Valence band** since it contains all the valence electrons. The upper band is called the **Conduction band** since excitation of electrons into this band is primarily responsible for electrical conduction.

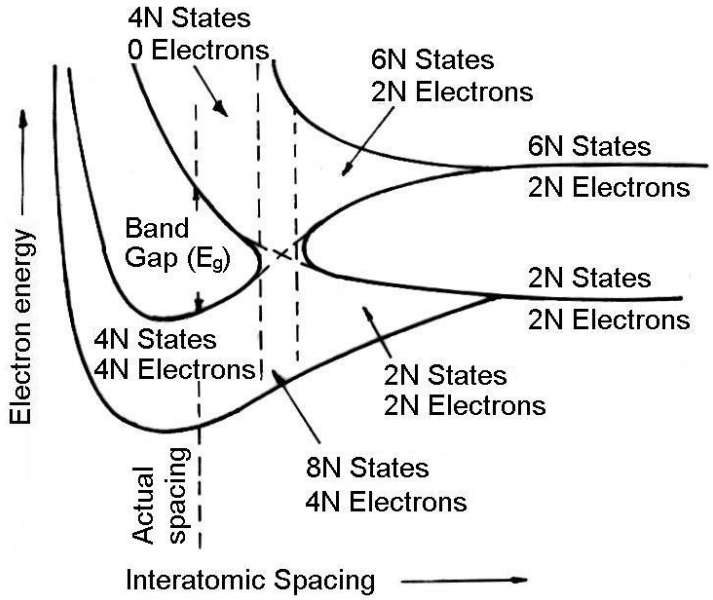


Fig. 1.5 Energy bands in a diamond crystal as a function of inter-atomic spacing

1.2.2 Metals, semiconductors and insulators

For electrical conduction, i.e. for electrons to experience acceleration in an applied electric field, they must be able to move into new energy states. That is, there must be empty (not already occupied) energy states available to electrons. For diamond, however, the valence band is almost completely filled and therefore hardly any empty states are available for electrons. On the other hand, there are very few electrons present in the conduction band and hence there is little possibility of charge transport. So the electrical conductivity is very poor and diamond is an insulator.

Semiconductors have identical energy band structures as insulators, i.e. a filled valence band and an empty conduction band at 0K. The only difference is in the size of the energy gap, which is smaller in semiconductors. For example, the band gap of diamond is 5 eV, while that of silicon is only 1.1 eV. The band gaps of some common semiconductors are given in Table 1.1. The relatively small separation between valence and conduction bands in semiconductors allows electrons to be excited from the filled valence band into the empty conduction band by thermal excitation at room temperature. This introduces electrons in an almost empty conduction band as well as vacant states in an almost full valence band, thereby increasing the conductivity of the material.

On the other hand, in the case of metals, bands either overlap or they are partially filled. Thus electrons have empty states available to them into which they can move under the influence of an electric field. This results in good electrical conductivity of most metals. The differences in the energy band structure of metals, semiconductors and insulators are shown in Fig 1.6.

The elements in the group IV of the periodic table present an interesting case. All these elements have diamond like structure in which the neighbouring atoms are bound by cohesive forces. As the atomic number increases, these cohesive forces weaken causing a reduction in the melting point as well as in the energy gap. Thus there is a transition from insulating (diamond) to semiconducting (silicon, germanium) to metallic (tin) behaviour as we consider successive elements under this group with increasing atomic number. The melting points and the band gaps of these elements along with their atomic numbers are presented in Table 1.2.

Semiconductor	**Bandgap (eV)**
Germanium (Ge)	0.66
Silicon (Si)	1.12
Indium Phosphide (InP)	1.35
Gallium Arsenide (GaAs)	1.42
Silicon Carbide (SiC)	2.99
Gallium Nitride (GaN)	3.36

Table 1.1 Bandgaps of some common semiconductors

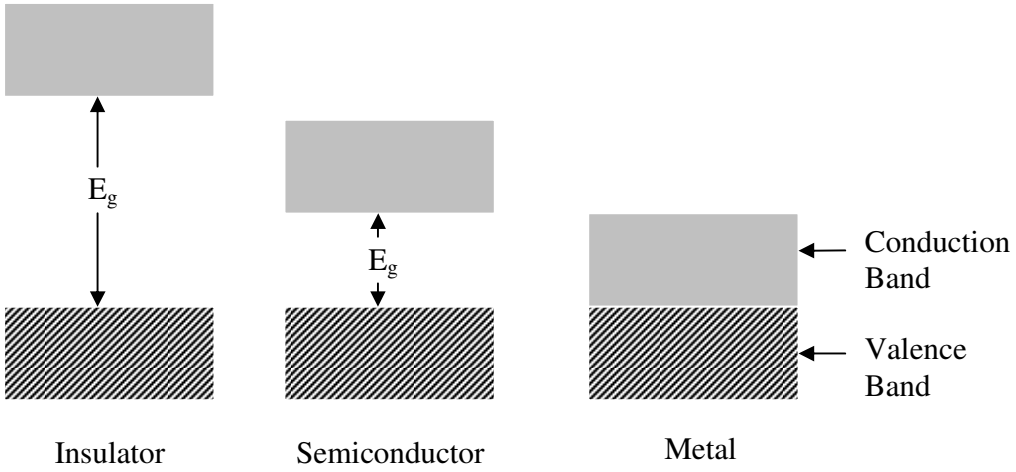


Fig.1.6 Typical band structures of insulator, semiconductor and metal

Element	Atomic Number	Bandgap (eV)	Melting Point (°C)
Carbon	6	5.30	3800
Silicon	14	1.12	1417
Germanium	32	0.66	937
Tin (α-Sn)	50	0.08	232

Table 1.2 Atomic number, bandgap and melting points of some Group IV elements

1.2.4 Charge carriers in Semiconductors - Electrons and Holes

As the temperature is raised from 0K, some electrons in the valence band receive enough thermal energy to be excited into the conduction band. So, now there are some electrons in the previously empty conduction band. Also, there are now some unoccupied states (holes) in the normally filled valence band. Thus by exciting an electron from the valence band to conduction band, an electron-hole pair (EHP) is created as represented schematically in Fig.1.8. After excitation into the conduction band, an electron is surrounded by a large number of unoccupied states. For example, the equilibrium number of EHP in pure silicon at 300K is $1.5 \times 10^{10} cm^{-3}$ compared to the silicon atom density of the order of 10^{22} cm^{-3}. So the few electrons in the conduction band are free to move about via the many available empty states. In all the following discussions, we shall concentrate on electrons in the conduction band and holes in the valence band. The movement of these two types of charge-carriers accounts for the current flow in a semiconductor.

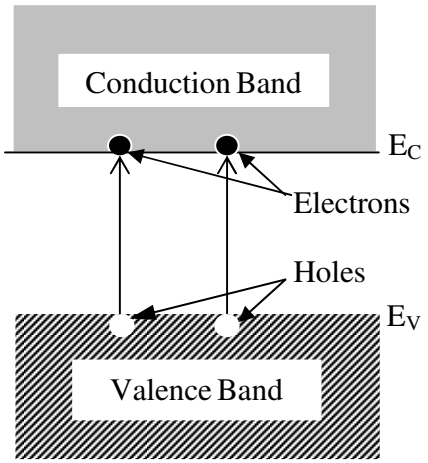


Fig. 1.8 Creation of an EHP when an electron is excited
from the valence to the conduction band

We can now relate to the discussion in Section 1.1 using the Chemical Bond model. At 0K, all valence electrons are tightly bound through the covalent bonds between neighbouring atoms. As the temperature increases, thermal vibration of the lattice can impart sufficient energy to some of the electrons allowing them to break the bonds and move freely inside the crystal. These free electrons can now participate in electrical conduction. The energy needed to break the bond is actually the band gap energy in the energy band diagram.

The energy band diagram actually reflects the electronic energy, i.e. if an electron gains energy it moves up in the energy band diagram. On the other hand, a hole, which gains energy, moves down due to its opposite charge. Thus holes seeking the lowest energy state are available at the top of the valence band (E_V) while electrons are available at the bottom of the conduction band (E_C). The unit of energy generally used is electron-Volt (eV), which is equal to 1.6×10^{-19} J. The name electron-Volt is due to the fact that if

an electron falls through a potential of 1V, the kinetic energy gained, which is equal to the decrease in its potential energy, is

$$E = qV = 1.6\text{x}10^{-19}\ \text{C x } 1\text{V} = 1.6\text{x}10^{-19}\ \text{J} = 1\ \text{eV} \quad (\because \text{Coulomb x Volt = Joule})$$

So in the energy band diagram if electron energy (E) goes up by 1eV, it means equivalently that it has fallen through a potential (V) of 1V. Note that although V and E are dimensionally different, they are numerically the same value. So, we may say that the energy band diagram also reflects what is known as the electronic potential, which is the negative of the electrostatic potential (the electrostatic potential is defined with respect to a positive charge). For an electron in the conduction band having an energy E, the energy corresponding to the bottom of the conduction band (E_C) is its potential energy, while ($E - E_C$) is its kinetic energy. This is shown in Fig. 1.9(a).

If we apply a voltage V across a semiconductor, the energy band diagram will be as shown in Fig.1.9(b). As the electron moves under the influence of the applied potential V, its loss in potential energy (qV) is equal to the gain in kinetic energy. However, as the electron moves, it suffers a number of collisions, giving up its kinetic energy to the lattice (generating heat), and reverting to the bottom of the conduction band. An equivalent analogy is that of a stone of mass 'm', when it is at a height 'h', has a potential energy equal to 'mgh', where 'g' is the acceleration due to gravity. When it falls and reaches the ground level under the influence of gravitational field, its entire potential energy is converted into kinetic energy. However, as it 'collides' with our planet, it ultimately loses all its kinetic energy and comes to rest.

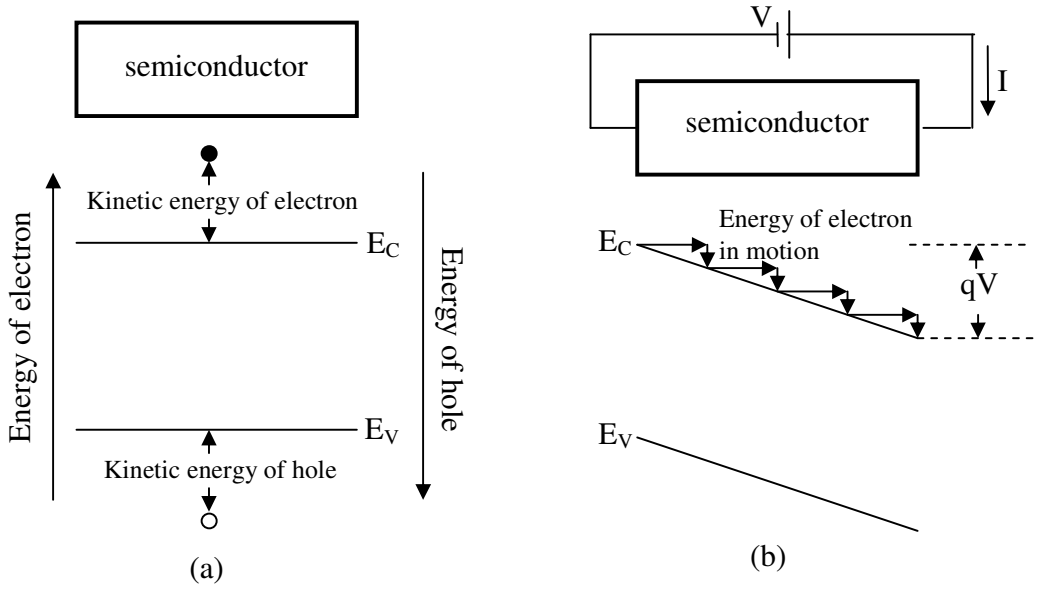


Fig. 1.9 Energy band diagram of a semiconductor under (a) zero bias and (b) under biasing condition. The energy of an electron in motion which loses energy due to collisions is also shown

1.2.5 Intrinsic and extrinsic semiconductors

A perfect semiconductor crystal with no impurities or defects is called an intrinsic semiconductor. In such a material, the valence band is filled and the conduction band is empty at 0K as already mentioned in Section 1.2.2. At higher temperature, EHPs are generated as the covalent bonds break and valence band electrons are excited to E_C as explained in Section 1.2.4. These EHPs are the only carriers in an intrinsic semiconductor. Since electrons and holes are created in pairs, the electron concentration in the conduction band (n) is equal to the hole concentration in the valence band (p), i.e.

$$n = p = n_i \qquad \text{……..(1.9)}$$

where **n_i is called the intrinsic carrier concentration**. As the temperature increases, lattice vibrations also increase and more EHPs are created. Thus n_i increases with temperature. Also, if the energy band gap is smaller, less energy is required for EHPs to

be generated. So at a given temperature, n_i will be higher for a material with smaller bandgap. For example, for silicon $n_i = 1.5x10^{10}$ cm^{-3} at 300K while at 500K it is about $3x10^{14}cm^{-3}$. On the other hand, GaAs which has a larger band gap than silicon has $n_i = 1.79x10^6$ cm^{-3} at 300K. The exact dependence of n_i on temperature and band gap energy will be derived subsequently.

We have already discussed in Section 1.1 that it is possible to create additional free electrons or holes by introducing a suitable dopant in the semiconductor. This concept will now be explained with the help of Energy Bands. When impurities are introduced in a perfect crystal, additional energy levels are created in the energy band structure. For example, a group V element (P, As and Sb) will introduce energy level E_D very close to E_C in silicon or germanium. This level is filled with electrons at 0K and with very little thermal energy these electrons can be excited into the conduction band. Such an impurity is therefore called a **donor** as it donates electrons. Thus, even when the temperature is low so that the number of thermally generated carriers is small, there is a large concentration of electrons in the conduction band of this type of doped semiconductor, i.e. $n >> n_i, p$. Such a semiconductor is called **n-type**. Again, if an element from group III is used to dope silicon or germanium, a level E_A is created very close to E_V. This level is empty at 0K but with little thermal energy, electrons from the valence band can be excited into this level, creating a large number of holes in the valence band. As this level accepts electrons, this type of dopants is termed an **acceptor** and the material is called **p-type**, where $p >> n_i, n$. Energy band diagrams of n-type and p-type semiconductors with the donor and acceptor levels are shown in Fig. 1.10. It may be mentioned here that the process of donating or accepting electrons ionizes the impurity

atoms, the donors becoming positively charged, while the acceptors become negatively charged as shown in Fig. 1.10. This will be further discussed in Section 1.3.7.

When a semiconductor is doped n-type or p-type, only one type of carriers dominates. These are called the **majority carriers**. Carriers of the other type, whose concentration is much less, are called the **minority carriers**. Thus electrons in n-type semiconductor and holes in p-type semiconductor are majority carriers while electrons in p-type semiconductor and holes in n-type semiconductor are minority carriers.

Q 1.1 Why is the intrinsic carrier concentration a constant and not an increasing function of time, inspite of EHPs being continuously generated at room temperature?

Till now we have only discussed the process of generation of EHPs, where an electron in the valence band gains sufficient energy to reach the conduction band. A simultaneous process called **recombination** also takes place in the semiconductor, where an electron in the conduction band gives up energy to go back to the valence band and recombines with a hole, thereby annihilating an EHP. The generation rate (G), defined as the number of EHPs generated per unit volume per second, at a given temperature would be more for lower band gap materials. Also, for a particular semiconductor G is higher at a higher temperature. On the other hand, the recombination rate (R), defined as the number of EHPs which recombine per unit volume per second, depends on the number of available carriers, and is actually proportional to the product of the number of free electrons and holes. If $G > R$, the number of carriers will increase as the number of carriers generated is more than the number of carriers recombining. On the other hand, if $G < R$, the number of carriers will reduce. An equilibrium will be reached and the number of carriers will become constant only when $G = R$. Now let us conduct a thought

experiment. Let the temperature of a semiconductor sample be raised instantaneously at t = 0 from 0K to 300K and maintained at 300K for t > 0. Let the generation rate at 300K be given by G_{300}. Initially, there are no carriers present, so R = 0. As G_{300} > R, the number of carriers increases and R keeps increasing with time. This process continues till G_{300} = R and the carrier concentration reaches an equilibrium value, say $(n_i)_{300}$. Now at $t = t_1$, the temperature of the semiconductor is suddenly raised to 400K. Let the generation rate at 400K be G_{400}. Since $G_{400} > G_{300}$, initially G > R, and the carrier concentration keeps increasing with a consequent increase in R. Again an equilibrium is reached when G = R, and let the equilibrium concentration at 400K be $(n_i)_{400}$. It is quite obvious that $(n_i)_{400} > (n_i)_{300}$. Thus the intrinsic carrier concentration increases with temperature. Similarly, we can also argue that the intrinsic carrier concentration will be lower for higher bandgap materials. We shall develop quantitative relations for the dependence of n_i on bandgap and temperature, as well as discuss recombination in more detail, later in this chapter.

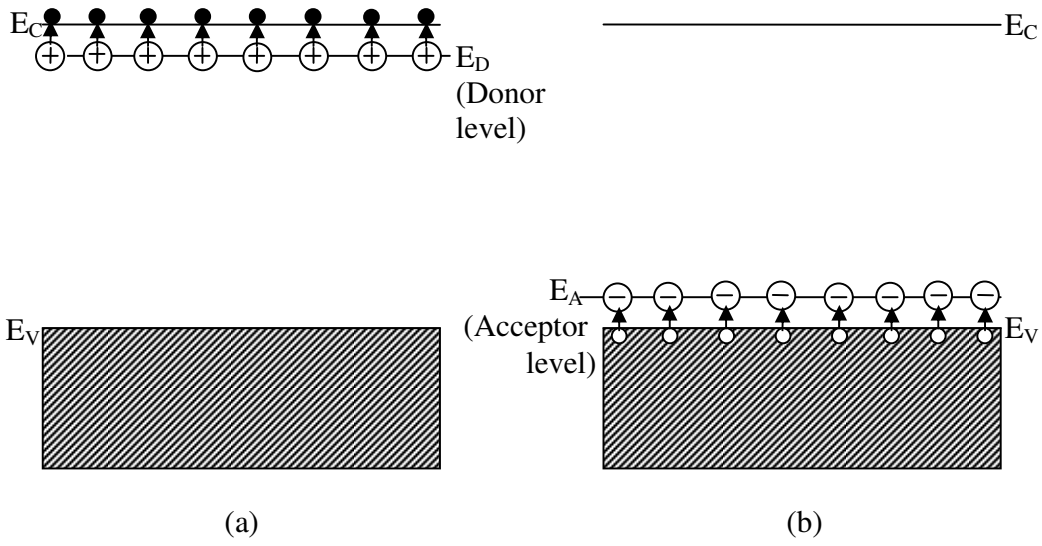


Fig. 1.10 Energy band diagrams of (a) n-type semiconductor with donor level E_D and (b) p-type semiconductor with acceptor level E_A. Ionization of the impurities have resulted in free electrons in the conduction band of the n-type semiconductor and holes in the valence band of the p-type semiconductor